#QUESTION 1

	Variable <chr>	Control_Mean <dbl>	Treated_Mean <dbl>	Difference <dbl>	P_Value <dbl>
1	numhh_list	1.1959945	1.2885549	0.092560355	0.0000000000000000000000000000000002494734
2	female	0.5686409	0.5627055	-0.005935381	0.5083839488080061919106356072006747126 6
3	age	40.6060606	40.9863786	0.380317975	0.0725383715643488635738123093688045628 4
4	race_white	0.6898387	0.6868322	-0.003006473	0.7203986276612390771134641909156925976 3
5	hs_degree	0.6823006	0.6795052	-0.002795337	0.7404890165493600573398680353420786559 6
6	college_degree	0.1116056	0.1153906	0.003785023	0.5099917633369303038293196550512220710 5
7	health_baseline	0.2716535	0.2703930	-0.001260558	0.8755275505972432981138808827381581068 0

#QUESTION 2

Answer: Based on the balance table in Part 1, the baseline characteristics **appear to be balanced** between the treatment and control groups. While [numhh_list] shows a p-value slightly below 0.05, given that we are testing multiple hypotheses (7 variables), it is statistically expected that 1 out of 20 tests might show significance purely by chance. Since the majority of variables are balanced, we can still conclude that the randomization was generally successful.

#QUESTION 3

	Variable <chr>	Control_Mean <dbl>	Treated_Mean <dbl>	Difference <dbl>	P_Value <dbl>
1	count_visit_dr	5.74586207	6.14155395	0.395691879	0.0669625946
2	count_visit_er	1.01323024	0.97734068	-0.035889564	0.3120524813
3	out_of_pocket_spend	563.45652987	490.41234601	-73.044183862	0.0008209446
4	health_score	0.08159156	0.08003334	-0.001558223	0.3103208248
5	happy	0.74562607	0.75799875	0.012372673	0.1138374639

5 rows

Answer: Based on the analysis of this sample, the Medicaid lottery proved most effective as a financial safety net rather than a driver of immediate health improvements. The most robust finding is a statistically significant reduction in out-of-pocket spending ($p < 0.05$), indicating that insurance successfully protected low-income individuals from the economic burden of medical costs. Regarding healthcare utilization, we find suggestive but not definitive evidence of an increase; doctor visits (visit_dr) showed a marginally significant increase ($p = 0.067$, significant at the 10% level), implying a trend toward utilizing more outpatient care when financial barriers are removed. However, these changes did not translate into statistically significant improvements in well-being or physical outcomes within the study period, as neither self-reported happiness (happy) nor objective health scores (health_score) showed significant differences between the treated and control groups ($p > 0.05$). This suggests that while Medicaid improves financial security, it does not automatically lead to measurable changes in physical health or happiness in the short term for this specific sample.

#QUESTION 4

```
[1] "=== Question 4: Regression Results ==="

Dependent Variable (Y): count_visit_dr
  -> Regression Beta1 (Diff): 0.3956919
  -> Regression P-value:      0.06696259

Dependent Variable (Y): count_visit_er
  -> Regression Beta1 (Diff): -0.03588956
  -> Regression P-value:      0.3120525

Dependent Variable (Y): out_of_pocket_spend
  -> Regression Beta1 (Diff): -73.04418
  -> Regression P-value:      0.0008209446

Dependent Variable (Y): health_score
  -> Regression Beta1 (Diff): -0.001558223
  -> Regression P-value:      0.3103208

Dependent Variable (Y): happy
  -> Regression Beta1 (Diff): 0.01237267
  -> Regression P-value:      0.1138375
```

Answer: the P-value in question 4 are equal to question 3

#QUESTION 5

We can't just take the results from Oregon and assume they apply perfectly to the entire U.S. for three simple reasons:

1. Different People: The people in this experiment actively signed up for the lottery, which means they probably needed insurance more than the average person. A national policy would cover everyone—including people who aren't motivated to sign up or who are healthier—so the results might look very different.

2. Different Scale: This experiment only added about 10,000 people, which the local doctors could handle. But expanding to the whole country involves millions of people. That huge surge in demand could overwhelm hospitals and cause long wait times (a "system congestion" issue) that this small experiment couldn't predict.

3. Different Place: Oregon isn't the whole U.S. Its demographics (like race or age) and local policies are unique. We can't simply assume that what worked in Oregon will work exactly the same way in states with very different populations, like Texas or New York.

#QUESTION 6

Answer: No, I wouldn't expect the results to be the same.

The big problem with just looking at real-world data (observational studies) is selection bias. In the real world, people aren't assigned insurance by a lottery; they choose it, or get it through their jobs. This creates two huge issues:

1. The "Wealthier is Healthier" Problem: People with insurance often have better jobs and more money. Because they are wealthier, they are usually healthier to begin with (better food, less stress). If we just compare them to uninsured people, insurance will look like a miracle cure, but we are really just seeing the effect of being wealthy. This would overestimate the benefits.

2. The "Sick People Buy Insurance" Problem (Adverse Selection): On the flip side, people who know they are sick are more likely to try to get insurance. This makes the "insured group" look sicker than the "uninsured group." This would underestimate the benefits.

Conclusion: The Oregon experiment is special because the lottery made the two groups exactly the same on average. Real-world data mixes up "the effect of insurance" with "the effect of being rich or sick," so you can't get a clean answer like you can in an RCT.

R

```
###QUESTION 1
```{r }
options(scipen = 999)
#define baseline
baseline_vars <- c("numhh_list", "female", "age", "race_white",
                   "hs_degree", "college_degree", "health_baseline")
#create an new frame to set outcome (row name)
balance_results <- data.frame(Variable = character(),
                              Control_Mean = numeric(),
                              Treated_Mean = numeric(),
                              Difference = numeric(),
                              P_Value = numeric(),
                              stringsAsFactors = FALSE)
#use for loop
for (col_name in baseline_vars) {
  control_data <- ohie_data[ohie_data$treated == 0, col_name]
  treated_data <- ohie_data[ohie_data$treated == 1, col_name]
  mean_c <- mean(control_data, na.rm = TRUE)
  mean_t <- mean(treated_data, na.rm = TRUE)
  diff_val <- mean_t - mean_c
  #t-test
  t_test_res <- t.test(treated_data, control_data, var.equal = TRUE)
  #p-value
  p_value <- t_test_res$p.value
  balance_results[nrow(balance_results) + 1, ] <- list(col_name, mean_c, mean_t, diff_val, p_value)
  print("=== Question 1: Balance Table ===")
  print(balance_results)
}
colnames(ohie_data)
```
```

```
###QUESTION 3
```{r}
#define variable name
outcome_vars <- c("count_visit_dr", "count_visit_er", "out_of_pocket_spend", "health_score", "happy")
#create an new frame to set outcome (row name)
outcome_results <- data.frame(Variable = character(),
                              Control_Mean = numeric(),
                              Treated_Mean = numeric(),
                              Difference = numeric(),
                              P_Value = numeric(),
                              stringsAsFactors = FALSE)
#calculate the for loop
for (col_name in outcome_vars) {
  control_data <- ohie_data[ohie_data$treated == 0, col_name]
  treated_data <- ohie_data[ohie_data$treated == 1, col_name]
  mean_c <- mean(control_data, na.rm = TRUE)
  mean_t <- mean(treated_data, na.rm = TRUE)
  diff_val <- mean_t - mean_c
  t_test_res <- t.test(treated_data, control_data, var.equal = TRUE)
  p_value <- t_test_res$p.value
  outcome_results[nrow(outcome_results) + 1, ] <- list(col_name, mean_c, mean_t, diff_val, p_value)
}
#print the outcome
print("=== Question 3: Outcomes Table ===")
print(outcome_results)
```
```

```
###QUESTION 4
```{r}
print("=== Question 4: Regression Results ===")
# use for loop
for (col_name in outcome_vars) {
  formula_str <- paste(col_name, "~ treated")
  model <- lm(as.formula(formula_str), data = ohie_data)
  model_summary <- summary(model)
  beta_1 <- model_summary$coefficients[2, 1]
  p_val_reg <- model_summary$coefficients[2, 4]
  cat("\nDependent Variable (Y):", col_name, "\n")
  cat("  -> Regression Beta1 (Diff):", beta_1, "\n")
  cat("  -> Regression P-value:      ", p_val_reg, "\n")
}
```
```